

Final Project & Literature Review Proposal

Group 17:

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Final Project Proposal: Terrain classification for NASA's mars2020 (Perseverance) rover

AI4Mars is a collection of open dataset of images taken by the various rovers currently operating on mars. The images are also label by citizen science initiatives through the *zooniverse* platform, with the final labels being verified by experts. The dataset we chose to work with is the navigation images of the Mars 2020 Perseverance rover.

The dataset consists of 3827 unique labeled images taken by the navigation cameras with different portions of the terrain labeled into one of 5 classes, according to table 1. The labels are written into the rgb colors of a label image. A reference of representative classes is show in figure 1, and an example of the label mask is shown in figure . This images are taken from a conference paper about the AI4Mars project [9] and are from the MSL Curiosity navigational dataset. Our dataset is more recent and from a different rover, but the classes and classification methods are the same.

The goal of our final project is to learn how to use the images provided by the AI4Mars data set to generate a design matrix, apply one or more machine learning method to segment the images in the 4 classes, and compare our results with other published results for martian terrain classification [1] [6].

Literature Review

Our literature review will entail studying two different sets of articles. The first are the articles related to the AI4mars project, detailed the dataset labeling process, as well as given an overview of the available data, and then assess the ability of different Machine Learning Models to do image segmentation on the images provided in the AI4Mars dataset or similar datasets[9][1][6][2][4][10].

The other set of papers we will study will be related to other image segmentation methods unrelated to the AI4Mars project, with the goal to inform which data processing techniques and methods we could apply in our final project [3][5][7][8].

Label	Type	RGB Label
0	soil	[0,0,0]
1	bedrock	[1,1,1]
2	sand	[2,2,2]
3	big rock	[3,3,3]
255	null / unlabeled	[255,255,255]

Table 1: Labels for the Mars2020 Navigation dataset

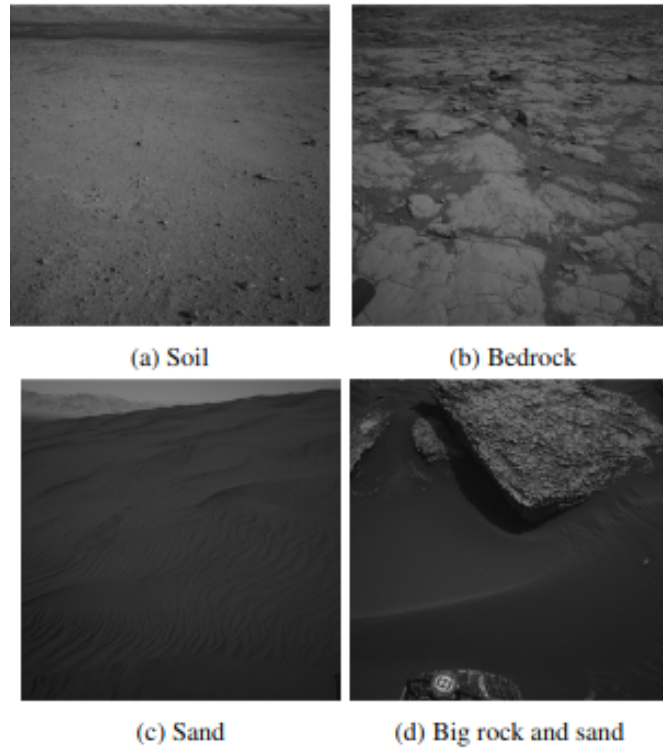


Figure 1: Representative examples of each class, taken from [9]

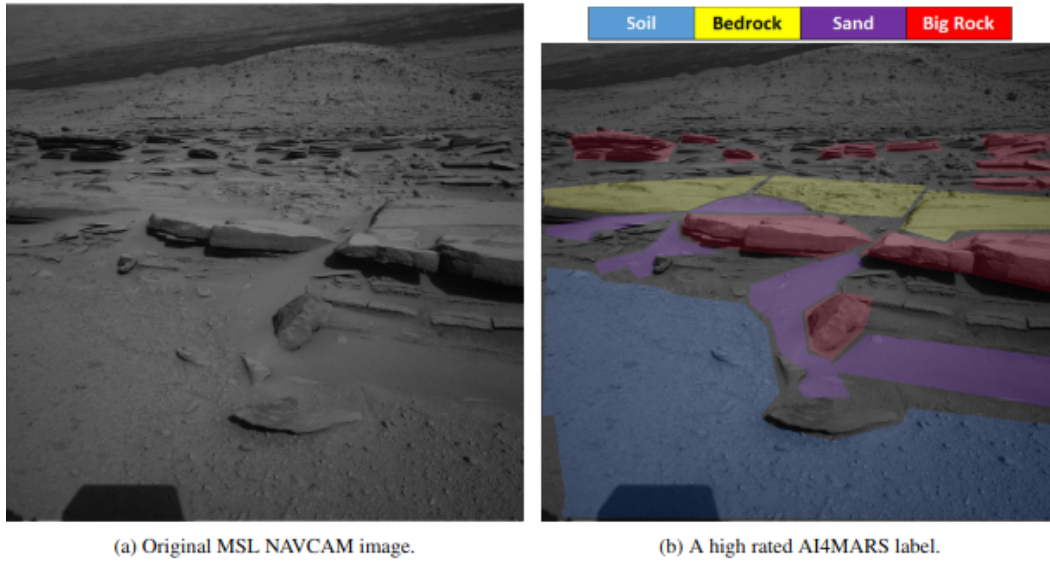


Figure 2: Example of original navigational camera image vs labeled image, taken from [9]

References

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